SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)
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॥वसुधैव कुटुम्बकम्॥

CSE - Academic Year 2026-27
Java Programming Lab | Batch 2025-29

ASSIGNMENT : 19

Name of Student	Anushree Shinde
PRN No.	25070122041
Batch	CSE A - 2
Class	Java Programming Lab
Academic Year & Semester	2025-29 : Semester 3
Date of Submission	

TITLE : Develop a Java application to establish connectivity with a database and perform data retrieval using the SELECT operation.

Learning Objectives : To establish database connectivity and retrieve records using JDBC SELECT statements.

Theory :

Java Database Connectivity (JDBC) is a standard Java API that enables Java applications to communicate with relational databases. It allows a Java program to connect to databases such as MySQL and execute SQL queries.

The **Connection** object establishes a connection with the database, while the **Statement** object is used to execute SQL statements. The **SELECT** statement retrieves records from a database table. The retrieved records are stored in a **ResultSet** object, which allows the data to be accessed row by row.

Exercise :

1. Create a Java application to display **student records from database** using SELECT query.

```
import java.sql.*;

public class java19a {
    public static void main(String[] args) {
        try {
            Connection con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/jdbc_lab",
                "root",
                "anushree"
            );

            System.out.println("Database successfully connected..");

            Statement st = con.createStatement();
            ResultSet rs = st.executeQuery("select * from student");

            System.out.println("Student Records");
            System.out.println("----------------------------");

            while (rs.next()) {
                System.out.println(
                    rs.getInt("roll_no") + "   " +
                    rs.getString("name") + "   " +
                    rs.getString("course") + "   " +
                    rs.getInt("marks")
                );
            }

            con.close();
        } catch (Exception e) {
            System.out.println(e);
        }
    }
}
```

In SQL :

```sql
create database jdbc_lab;
use jdbc_lab;

create table student (
    roll_no int,
    name varchar(50),
    course varchar(50),
    marks int
);

insert into student values
(101, 'Anushree', 'CSE', 95),
(102, 'Anwesha', 'CSE', 82),
(103, 'Natasha', 'IT', 91);

select * from student;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

roll_no	name	course	marks
101	Anushree	CSE	95
102	Anwesha	CSE	82
103	Natasha	IT	91

OUTPUT :

```
Database successfully connected..
Student Records
-----------------------------
101  Anushree  CSE  95
102  Anwesha  CSE  82
103  Natasha  IT  91
```

2. Create a Java application that connects to a database and displays **product details** using a **SELECT query**. Show product ID, product name, quantity, and price.

```
import java.sql.*;

public class java19b {
    public static void main(String[] args) {

        try {
            Connection con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/jdbc_lab",
                "root",
                "anushree"
            );

            System.out.println("Database successfully connected..");

            Statement st = con.createStatement();
            ResultSet rs = st.executeQuery("select * from product");

            System.out.println("Product Details");
            System.out.println("------------------------------");

            while (rs.next()) {
                System.out.println(
                    rs.getInt("product_id") + "  " +
                    rs.getString("product_name") + "  " +
                    rs.getInt("quantity") + "  " +
                    rs.getDouble("price")
                );
            }

            con.close();

        } catch (Exception e) {
            System.out.println(e);
        }
    }
}
```

In SQL :

```sql
create table product (
    product_id int,
    product_name varchar(50),
    quantity int,
    price double
);

insert into product values
(1, 'Laptop', 10, 65000),
(2, 'Keyboard', 25, 1200),
(3, 'Mouse', 30, 700);

select * from product;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

product_id	product_name	quantity	price
1	Laptop	10	65000
2	Keyboard	25	1200
3	Mouse	30	700

OUTPUT :

```
Database successfully connected..
Product Details
-----------------------------
1  Laptop  10  65000.0
2  Keyboard  25  1200.0
3  Mouse  30  700.0
PS C:\Users\softlink\Desktop\java code>
```

Outcome :

- Understood database connectivity using JDBC.
- Learned to establish a connection between Java and MySQL.
- Implemented SELECT queries to retrieve records from database tables.
- Learned to access retrieved records using the ResultSet object.

Conclusion :

This assignment demonstrated database connectivity in Java using JDBC. The student and product records were successfully retrieved from the database using SELECT queries and displayed using the **ResultSet** object. This helped in understanding the basic process of retrieving database records using Java.